RIGA NORDIC UNIVERSITY
of Applied Sciences

JAVASCRIPT COURSE

Make your websites dynamic and interactive with JavaScript

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COURSE OVERVIEW

- Make your websites dynamic and interactive
- Create features and stand-alone applications
- Learn to manipulate the DOM
- Apply object-oriented programming principles
- Fetch real-world data using APIs
- Major sections:
  - Organizing Your JavaScript Code
  - JavaScript in the Real World
  - Asynchronous JavaScript and APIs
  - Testing JavaScript
  - A Bit of Computer Science
  - Intermediate Git
  - Finishing Up

Organizing Your JavaScript Code (Part 1)

- Organizing Code with Objects
- Object Constructors
- **Project: Library** – build a book-tracking app
- Factory Functions and the Module Pattern
- **Project: Tic Tac Toe** – a playable game
- Classes
- ES6 Modules

Organizing Your JavaScript Code (Part 2)

- npm – the Node package manager
- Webpack – bundling your code for the browser
- **Project: Restaurant Page** – a multi-page site built with Webpack
- Revisiting Webpack
- JSON – data format
- OOP Principles
- **Project: Todo List** – a task manager

JavaScript in the Real World

- Linting – catching errors and enforcing style
- Form Validation with JavaScript
- ECMAScript – the language standard and its versions

Asynchronous JavaScript and APIs

- Asynchronous Code – callbacks, promises
- Working with APIs – fetching real-world data
- Async and Await – modern asynchronous syntax
- **Project: Weather App** – fetch and display live weather data

TESTING JAVASCRIPT

- Testing Basics – why and how to test
- **Project: Testing Practice** – write your first tests
- More Testing – deeper testing techniques

A BIT OF COMPUTER SCIENCE

- A Very Brief Intro to CS
- Recursive Methods
- **Project: Recursion**
- Time Complexity and Space Complexity
- Common Data Structures and Algorithms
- **Project: Linked Lists** and **Project: HashMap**
- **Project: Binary Search Trees**
- **Project: Knights Travails**

INTERMEDIATE GIT

- A Deeper Look at Git – branching, merging, history
- Working with Remotes – collaborating via GitHub
- Using Git in the Real World – team workflows

Finishing up with JavaScript

- **Project: Battleship** – the capstone project
- Conclusion – bringing it all together

Key Takeaways

- Write well-organized JavaScript with objects, factories, and classes
- Use npm, Webpack, and modules like a professional
- Fetch real-world data with APIs and async/await
- Test your code for reliability
- Apply computer science fundamentals to solve problems